

One designer. Three roles that normally require separate teams.

What a properly planned, methodically executed
AI-assisted product pipeline actually looks like.

Product Management · Product Design · Development

Vanguard's portal separates the data you need to decide from the tools you need to act.

VANGUARD PERSONAL INVESTOR PORTAL

- Holdings & balances in Portfolio tab
- Sell transactions in Transact tab
- No integrated tax impact modeling
- Lot-level selling requires navigating away from portfolio context
- No rebalancing + transaction workflow

HYPOTHETICAL TOOL

- Holdings & transactions unified
- Real-time tax impact modeling
- Automated MinTax recommendation
- Self-directed portfolio rebalancing

Not affiliated with Vanguard — portfolio concept only

Understanding all three roles makes me a better designer.

PRODUCT MANAGEMENT	PRODUCT DESIGN	DEVELOPMENT
Problem statement	33-component design system extracted from live portal	React 18 · TypeScript Zustand · Vite
11 functional requirements	12 annotated screens	Dual-objective optimization engine (tax + rebalance)
80-issue Jira backlog via Atlassian Rovo MCP	50-connection Figma prototype via Claude Code + Figma MCP	58/58 unit tests passing
Canonical sample dataset 43 lots · 3 accounts · 7 funds	PDB decisions log written to Notion via MCP	2/2 Playwright E2E flows
Multi-tool AI comparison Claude · ChatGPT · Gemini	4-stage progressive disclosure architecture	Deployed to Vercel GitHub Actions CI gate

~413 Claude AI prompts	~185 screenshots uploaded	~42 Claude Code prompts	5 major chat sessions	734-line CLAUDE.md spec	6 AI tools used
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The execution speed is real.

The quality depends on the quality of the spec.

✓ Design system extraction

Read Vanguard's live portal DevTools output and instantiated a 33-component Figma library — colors, typography, spacing, and component patterns — in a single session.

✓ Prototype wiring at scale

Wired 50 Figma prototype connections from a structured 3-prompt spec (DS audit → frame inventory → connection spec) with zero failures.

✓ Application scaffold

Built the full React application structure, routing, state management, and optimization engine from a structured CLAUDE.md handoff document.

✓ Cross-role document generation

Used MCP connections to write directly to Notion and Jira — updating requirements, creating 80 stories, and compiling documentation without copy-paste overhead.

AI will reassure you before it admits a gap.

The data quality problem

I asked Claude to generate rich sample data at the very start of the project — 43 lots, multiple account types, full tax lot detail. It produced data that looked complete and repeatedly confirmed the business logic was testing correctly.

It wasn't. Lot-level detail was fabricated to appear consistent. Errors only surfaced later during development, requiring significant rework to the canonical dataset and compensating changes across the codebase.

Verify independently

Treat AI's self-reported test results as a starting point, not a conclusion

Data first

Demand and verify the full canonical dataset before any design or dev work begins

Silent failure is the real risk

AI doesn't say 'I can't do this.' It produces something plausible and moves on

The takeaway: AI doesn't eliminate human judgment. It relocates it — from generating outputs to verifying them.

The live app has subtle coach marks that guide first-time users without distracting repeat users.

vanguard-ai-pipeline-jade.vercel.app

- 1 Enter \$25,000 as the withdrawal amount
- 2 Click "Get Recommendation" — see the AI optimization engine output
- 3 Review the Summary Banner (ST gains, LT gains, estimated net tax) and submit the order, or continue with more sophisticated features...
- 4 Switch to Manual mode — try changing a fund to SpecID cost basis
- 5 Open lot details on VTSAX — explore lot-level tax efficiency
- 6 Navigate to Scenario Analysis — compare scenarios side by side

This is the third in a three-part series.

01 — AI-Assisted Design

Feb 2026

Research → data model → design system → hi-fi wireframes → interactive prototype. All AI-assisted, all documented.

[View Prototype →](#) | [Portfolio →](#) | [Case Study →](#)

02 — AI-Assisted Development

Mar 2026

Design artifacts → deployed full-stack application. Next.js frontend, FastAPI backend, Claude Code autonomous builds.

[View Live App →](#) | [Portfolio →](#) | [Case Study →](#)

03 — AI Pipeline

Apr–Jul 2026

Complete PM → PD → Dev pipeline. PRD, Jira, design system, prototype, React app. One person, three roles.

[View Live App →](#) | [Portfolio →](#) | [Case Study →](#)

Full case study · live application · GitHub repo · Notion workspaces — all linked in the comments below.